

Traveling Salesman Problem (TSP)

Aim

To find the shortest possible route that visits each city exactly once and returns to the starting city using Python.

Objectives

Understand the Traveling Salesman Problem.

Find the minimum-cost tour.

Visit every city exactly once.

Return to the starting city.

Algorithm

1. Define the distance matrix between cities.
2. Generate all possible city tours.
3. Calculate the total distance for each tour.
4. Compare all tour distances.
5. Select the tour with the minimum distance.
6. Display the shortest path and its cost.

Pseudocode

```
START
Read distance matrix
Generate all possible tours
For each tour
    Calculate total distance
    If distance is minimum
        Store tour
Display shortest tour and distance
STOP
```

Python Code

```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

cities = [0, 1, 2, 3]
min_cost = float('inf')
best_path = None

for path in permutations(cities[1:]):
    tour = (0,) + path
    cost = 0

    for i in range(len(tour)-1):
        cost += graph[tour[i]][tour[i+1]]

    cost += graph[tour[-1]][0]

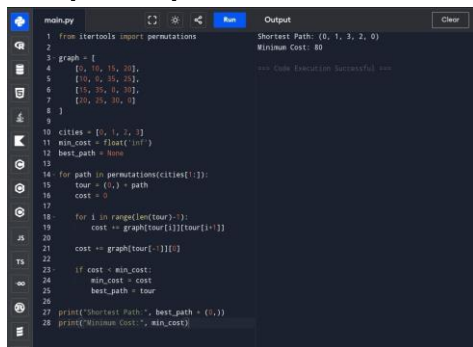
    if cost < min_cost:
        min_cost = cost
        best_path = tour
```

```
print("Shortest Path:", best_path + (0,))
print("Minimum Cost:", min_cost)
```

Result

The shortest tour and its minimum travel cost were found successfully.

Sample Output



```
1 from itertools import permutations
2
3 graph = [
4     [7, 10, 15, 20],
5     [10, 6, 30, 20],
6     [15, 30, 2, 30],
7     [10, 30, 30, 3]
8 ]
9
10 cities = [0, 1, 2, 3]
11 min_cost = float('inf')
12 best_path = None
13
14 for path in permutations(cities[1:]):
15     tour = [0,] + path
16     cost = 0
17     for i in range(len(tour)-1):
18         cost += graph[tour[i]][tour[i+1]]
19     cost += graph[tour[-1]][tour[0]]
20
21     if cost < min_cost:
22         min_cost = cost
23         best_path = tour
24
25 print("Shortest Path:", best_path + (0,))
26 print("Minimum Cost:", min_cost)
```

Shortest Path: (0, 1, 3, 2, 0)
Minimum Cost: 80

Flowchart

